

Ramiro Barcala Roca

Junior Software Developer / Systems Technician

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ABOUT ME

4th-year student at FIUBA pursuing a dual degree in Systems Analysis and Computer Engineering. Passionate about low-level software development, distributed architectures, and computer organization. I combine a solid hardware foundation in electronics with strong analytical skills to engineer robust, concurrent, and high-performance software.

EDUCATION

B.S. in Systems Analysis & Computer Engineering 2023 – Present
Faculty of Engineering, University of Buenos Aires (FIUBA) — Dual Degree Program (4th Year, 18 Passed Subjects)

Core courses: Algorithms & Data Structures (I, II, III), C++ Programming Workshop, Computer Organization, Computer Networks, Data Management.

Electronics Technician 2017 – 2022

Escuela Cristiana Evangélica Argentina (ECEA)

6-year technical degree: PCB modeling/fabrication, digital logic, microcontrollers, and SMD/THT precision soldering.

WORK EXPERIENCE

Repair & Hardware Technician 2022 – 2023 (1 year)
Sitek Electrónica SRL

- **Hardware Diagnosis:** Systematic fault detection and repair in industrial microelectronic components.
- **Industrial Maintenance:** Reconditioning of PLC switched-mode power supplies and execution of preventive maintenance routines.
- **Circuit Design:** Modeling, simulation, and manufacturing of custom printed circuit boards (PCBs).

FEATURED PROJECTS

Argentum Online — Client/Server Engine C++17 • TCP Sockets • Multithreading • SDL2

Real-time multiplayer client-server engine. Designed non-blocking TCP socket layers, custom binary protocol serialization, multithreaded synchronization using mutexes, and software design patterns (MVC, State, Observer).

Data Research & Predictive Modeling Python • Pandas • Scikit-Learn • EDA

Exploratory Data Analysis (EDA) pipelines and machine learning predictive models across weather, music streaming, and transit datasets. Evaluated through classification and regression metrics.

Network Protocol Analysis & Diagnostics (DNS & ICMP) Linux • Wireshark • DNS • Bash • LaTeX

Deep packet inspection and analysis of iterative vs. recursive name resolution using `dig` and Wireshark. Authored formal technical documentation in LaTeX.

SOFTWARE & SYSTEMS

Languages: C, C++, Rust, Python, Java, Assembler, Clojure, Smalltalk.

Networking & OS: Linux (Ubuntu/Debian), TCP/IP, Sockets, Wireshark, Bash, Git.

HARDWARE & LANGUAGES

Electronics: PCB Design/Routing, PLCs, AutoCAD, FreeCAD, LaTeX.

Languages: Spanish (Native), English Advanced B2 (FCE Oxford College, Grade B).